SOOHYUN CHOI

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
Hanyang University · Information and Intelligence Systems Lab (IISL) · Seoul, Republic of Korea
petersun0221@hanyang.ac.kr | schoish.github.io | github.com/SChoish

RESEARCH INTERESTS

Reinforcement learning and sequential decision making, with emphasis on iterative policy improvement, offline goal-conditioned RL, and generative long-horizon state-space planning.

EDUCATION

Hanyang University Sep. 2024–Present

Integrated M.S./Ph.D. Program in Electronic Engineering
Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong · GPA: 4.45/4.50

Hanyang University 2018–2024

B.S. in Electronic Engineering · Summa Cum Laude

MANUSCRIPT

Soohyun Choi*, Seonvin Cho*, and Songnam Hong. "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning." *Under review*, 2026. *Equal contribution. [Code]

- Developed a hierarchical offline GCRL framework that links subgoal selection to short-horizon execution through state-space bridges, inverse dynamics, and execution-time replanning.

SELECTED RESEARCH

Multi-step Proximal Policy Improvement 2026–Present

- Study re-centered, critic-guided proximal refinement and how effective step size and refinement depth govern stability across policy-improvement budgets and critic regimes. [Code]

PathFlower - Goal-Conditioned State Flows Ongoing

- Extend PathBridger toward final-goal-conditioned generative state flows that produce useful intermediate states for long-horizon offline control.

ALARM - Analog Circuit Optimization on a Riemannian Manifold 2024

- Formulated designer-selected performance equivalence classes and quotient geometry for circuit search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study. [Report]

REACT - Recursive Algorithm for RC Network Transfer Functions 2024–2025

- Developed a recursive closed-form transfer-function method for arbitrary RC trees and a dominant-pole reduced-order approximation validated against SPICE; SK hynix–Hanyang University sponsored project.

HONORS & FUNDING

- Top Excellence Award (Hanyang Electronics and Information Communications Alumni Association Special Award) - KRW 1,000,000, 2023
- Top Excellence Award, 15th Capstone Design Fair - KRW 1,000,000, 2023
- Graduation Honors, Hanyang University, 2024
- BK Research Assistantship (BK-RA), Hanyang University, 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I